

TIPAN BASEL

AI and AI tool enthusiast

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PROFESSIONAL SUMMARY

Computer Engineering student with a strong interest in Artificial Intelligence, Machine Learning, and software development. Experienced in building practical AI applications using Python and deep learning techniques. Enjoy solving real-world problems through hands-on projects while continuously improving technical and problem-solving skills. Seeking opportunities to contribute to innovative projects and grow as an AI Engineer.

EDUCATION

Advanced College of Engineering and Management | Kathmandu, Nepal Apr 2022 - May 2026
Bachelor in Computer Engineering Percentage : 72%
Participated on different events like Robotics, Extracourses and Programs.

PROJECTS

GovSathi:Nepali AI assistant for Government Services | 🐙 Github Apr 2025 - Feb 2026
Leader Developer

- Developed an AI-powered Nepali legal assistant using RAG to provide answers from 2 major government legal documents.
- Built a FAISS-based retrieval system with 1,000+ semantic chunks for context-aware information retrieval.
- Integrated RAG, voice processing, and FastAPI to create an end-to-end government-service assistance system.

Technologies / Tools Used : React, RAG, Whisper GTSS, MySQL

Diabetic Retinopathy Detection System | 🐙 Github Nov 2024 - Feb 2025
Leader Developer

- Trained a ResNet101 model on 3,672 retinal images across 5 diabetic retinopathy classes.
- Achieved approximately 85% classification accuracy using transfer learning and image augmentation techniques.
- Applied preprocessing and data augmentation techniques to improve model generalization on imbalanced retinal image data.

Technologies / Tools Used : Tensorflow, ResNet101, Python, OpenCV

Movie Recommender System Feb 2023 - Aug 2024
Individual Project

- Processed a movie dataset containing [X]+ movies to develop a personalized recommendation system.
- Implemented similarity-based recommendation techniques to generate relevant movie suggestions based on user preferences.

Technologies / Tools Used : Python, Pandas, NumPy, Scikit-learn, Jupyter Notebook, CSV Dataset

Library Management System Sep 2022 - Oct 2023
Individual Project

- Developed a web-based system for managing books, members, and borrowing and return records.
- Integrated MySQL for storing and managing library data efficiently.
- Created interactive web interfaces to simplify common library management tasks.

Technologies / Tools Used : HTML, CSS, JavaScript, PHP, MySQL, XAMPP

SKILLS

Programming Languages : C, C++, Python, JavaScript, PHP

Frameworks & Libraries : React, Next.js, FastAPI, MATLAB, Latex

Tools & Platforms : GitHub

Databases : MySQL, MongoDB

Soft Skills : Adaptability, Critical Thinking, Problem Solving, Time Management